Software Requirements Specification (SRS) Template

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1.0	18-08-2025	Instructor	SRS with diagrams embedded	

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for an Admin Interface to create and manage Question–Answer pairs with a user-facing search feature. It defines functional and non-functional requirements, interfaces, and constraints, serving as a reference for development, testing, and evaluation.

1.2 Scope

The system enables administrators to create, update, delete, and organize Question–Answer pairs through a dashboard, while end users can search and retrieve answers ranked by keyword relevance. It integrates full-text indexing for efficient search, CRUD APIs for content management, and a responsive front-end. External AI-based semantic search and third-party knowledge base integration are excluded from scope.

1.3 Audience

Developers, QA Engineers, System Integrators, Administrators, Content Managers, and End Users.

1.4 Definitions

List of acronyms: CRUD, API, UI, Q/A, RT, DB, JSON, HTTPS

2. Overall description

2.1 Product perspective

The system is a web-based platform that allows administrators to manage Question–Answer pairs and provides end users with a search interface. It includes a responsive front-end, CRUD APIs for data management, and a full-text indexing library for keyword-based ranking.

2.2 Major product functions (detailed)

- Create, update, and delete Question–Answer pairs
- Full-text keyword search with relevance ranking
- Display of suggested answers with highlighted keywords
- Feedback option for answer usefulness
- Automatic re-indexing of updated content- Role-based authentication and access control
- API endpoints for external integration

2.3 User roles and characteristics (expanded)

- **Admin**: maintains Question–Answer pairs, manages indexing, expects a clear dashboard and efficient workflows.
- **End User**: searches for answers, expects fast and accurate results

with minimal effort.
- **System Integrator**: connects external systems via APIs, expects well-documented endpoints.

2.4 Operating environment

Web-based system running on standard browsers (Chrome, Firefox, Edge), backend server with database and indexing engine, secure network (HTTPS), and responsive design across desktops, tablets, and mobile devices.

2.5 Constraints

Compliance with data protection standards (e.g., GDPR if applicable), response time for search queries under two seconds in normal load, APIs must follow REST principles, and mobile responsiveness must be ensured.

3. External interface requirements

(Replaces the ATM-specific content in Section 3.)

3.1 User interfaces

- **Admin Web UI (desktop-first, responsive):** dashboard with list/table of Q/A pairs; create/edit forms (rich text for answers), tag/labels editor, status (draft/published/archived), bulk actions (import/export), re-index button, and analytics cards (views, searches, CTR).

- **End-User Search UI (responsive):** single search box with instant results; keyword highlighting in snippets; sort by relevance/recency; filters (tags, date, author); pagination/infinite scroll; “Was this helpful?” feedback; related queries.

- **Accessibility:** keyboard navigation, ARIA roles, focus indicators, screen-reader labels, and high-contrast mode (WCAG 2.1 AA).

- **Internationalization (optional):** UI strings externalized; content language shown, with filter by language.

3.2 Hardware interfaces

- **None special.** Standard web client on desktop/mobile; backend runs on commodity servers or cloud VM/containers. (No card readers, cash dispensers, etc.)

3.3 Software interfaces

- **CRUD REST API (JSON over HTTPS):**
    - `POST /api/qa` create Q/A
    - `GET /api/qa` list with filters/sort
    - `GET /api/qa/{id}` retrieve
    - `PUT /api/qa/{id}` update
    - `DELETE /api/qa/{id}` soft-delete/archive
    - `POST /api/qa/bulk-import` CSV/JSON import
    - `POST /api/qa/{id}/publish` state change
- **Search API / Indexer:** integration with an indexing library/engine (e.g., Lucene/Elasticsearch/Meilisearch) via SDK or HTTP:
    - `POST /api/search/reindex` full/partial reindex
    - `GET /api/search?q=...&filters=...` ranked results with highlights
- **Auth:** OAuth2/OIDC or email+password; JWT access tokens; RBAC roles (`admin`, `editor`, `viewer`).
- **Telemetry & Logs:** structured logs (JSON), metrics endpoint `/metrics` (Prometheus format) and webhooks for events (publish, delete, reindex).
- **Notifications (optional):** SMTP or provider API for admin alerts (failed imports, index errors).

3.4 Communications

- **Transport:** TLS 1.2+; HSTS; strong ciphers.
- **API Policy:** CORS (allow configured origins), rate limiting, pagination, ETags.
- **Reliability:** exponential backoff/retries for index updates; idempotency keys on write endpoints.

- **Data formats:** UTF-8 JSON; CSV for bulk import; OpenAPI spec published at `/openapi.json`.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow QA-F-###.

4.1 Authentication & RBAC

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
QA-F-001	The system shall authenticate users and issue JWT access tokens	Functional	High	Security / Admin	Valid credentials → 200 + JWT; invalid → 401; token expiry honored. *Test:* TC-AUTH-01.	Requires JWT library, secure password hashing, DB for users, TLS; clarify password policy and MFA support.
QA-F-002	The system shall enforce RBAC with roles admin,editor, viewer	Functional	High	PCI-DSS	`editor` can CRUD Q/A except delete; `viewer` read-only; `admin` full access. *Test:* TC-AUTH-02.	Requires role-permission mapping, authorization middleware, audit logging; clarify if roles are fixed or

						configurable and hide unauthorized UI actions.
QA-F-003	The system shall support SSO via OAuth2/OIDC (configurable).	Functional	Medium	Security	When enabled, user can log in via provider; roles mapped from claims. *Test:* TC-AUTH-03.	Requires OAuth2/OIDC library, IdP availability, role-mapping config, local fallback; specify supported IdPs and role mapping rules.

4.2 Q/A Content Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria/Test case ref	Comments/Dependencies
QA-F-010	Create Q/A with fields: question (plain text), answer (rich text/Markdown), tags[], status, owner, language, and	Functional	High	Admin, Editor, End Users	POST with required fields creates record; returns 201 + ID. *Test:* TC-QA-01.	Requires CRUD API, DB schema, and validation; ensure mandatory fields and language support.

	SEO fields (slug, keywords).					
QA-F-011	Update Q/A fields and auto-version on save (keep last N versions, configura ble	Function al	High	Admin, Editor	PUT updates content; version history shows diff; restore works. *Test:* TC-QA-02.	Requires versionin g in DB and audit logging; ensure rollback/r estore of previous versions.
QA-F-012	Soft delete/ar chive Q/A; prevent end-user visibility when archived.	Function al	Medium	Admin	Archived item excluded from search/lis t (user side). *Test:* TC-QA-03.	Requires status field + search index exclusion ; consider GDPR permane nt delete complian ce.
QA-F-013	Bulk import Q/A from CSV/JSO N with validation report.	Function al	Medium	Admin	Valid rows created; invalid rows reported with line numbers and reasons. *Test:* TC-QA-04.	Requires file upload + parser/va lidator; define max file size and error reporting limits.
QA-F-	Tag/Cate	Function	Medium	Admin,	Tag	Requires

014	gory manage ment (create/re name/me rge tags).	al		Editor	merge reassigns all Q/A and updates index. *Test:* TC-QA-05.	tag DB + reindex logic; prevent orphan tags and ensure tag merge consisten cy
QA-F-015	Publish workflow: draft → in review → published (configur able).	Function al	High	Admin, Editor	*Acceptan ce:* Only published items appear in end-user search. *Test:* TC-QA-06.	Requires workflow state field + role enforcem ent; clarify if “Review” step is mandator y

4.3 Text Search and Ranking

Req id	Require ment	type	priority	Source/ Stakehol der	Accepta nce criteria/T est case ref	Comme nt/Depe ndencies
QA-F-020	Index question , answer, tags, and metadat a with field boosts (questio n > answer	Function al	High	Admin, End Users, System Architect	Index contains fields with configur ed boosts; verify via scoring debug. *Test:* TC-SRCH-	Requires indexing engine (Elastics earch/Lu cene), schema with boosted fields; ensure boosts are

	> tags).				01.	configurable.
QA-F-021	Keyword search shall return results ranked by relevance (TF-IDF/BM25 or engine default).	Functional	High	End Users, Admin	Queries match expected top results; deterministic within engine tolerance. *Test:* TC-SRCH-02	Requires search API integration and ranking algorithm; ensure deterministic ranking with debug tools.
QA-F-022	Support phrase queries, boolean operators, and filters (tags, date range, language, status=published).	Functional	Medium	End Users, Admin	`tag:billing "reset password"` narrows results correctly. *Test:* TC-SRCH-03.	Requires advanced query parser in index engine; ensure proper filter validation to avoid errors.
QA-F-023	Highlight matched terms in snippets with ellipses	Functional	Medium	End Users	Returned fragments show `<mark>` (or	Requires snippet generator + HTML sanitizer;

	and safe HTML.				equivalent) wrappers; no XSS. *Test:* TC-SRCH-04.	ensure safe rendering in UI.
QA-F-024	Support typo-tolerance (edit distance ≤1) and stemming (configurable per language).	Functional	Medium	End Users	"acount" → "account"; plurals normalize; can be disabled. *Test:* TC-SRCH-05.	Requires fuzzy search support in engine; ensure stemming is language-aware and configurable.
QA-F-025	Reindex on create/update/delete within 2 seconds (event-driven), plus manual full reindex.	Functional	High	Admin, DevOps	Changes visible in search within SLA; manual reindex endpoint works. *Test:* TC-SRCH-06.	Requires event-driven sync between DB and index; ensure fallback manual reindex API.

4.4 Suggested answers and feedback

Req ID	Requirements	Type	Priority	Source/ Stakeholders	Acceptance criteria/Test case	Comments/Dependencies

					ref	
QA-F-030	Show top N suggested answers as user types (autocomplete).	Functional	High	End Users, Admin	<150 ms median for prefix queries; suggestions click-through navigates to answer. *Test:* TC-SUG-01.	Requires search engine with prefix search + caching; ensure response time SLA is met
QA-F-031	Capture per-result feedback: helpful/not helpful, and click-through.	Functional	Medium	End Users, Admin, Analytics Team	Events recorded with result ID, query, user/session; visible in analytics. *Test:* TC-SUG-02.	Requires feedback API + analytics DB; ensure feedback is linked to session/user for insights
QA-F-032	Optional) Use implicit feedback to adjust boosts (learning-to-rank lite).	Functional	Low	Admin, Analytics Team	Items with high CTR get small boost (bounded); audit trail retained. *Test:* TC-SUG-03.	Requires learning-to-rank module + feedback data integration; ensure boosts are bounded to avoid bias.

4.5 End -User Search UI

Req ID	Requirements	Type	Priority	Source/ Stakeholder	Acceptance/Test case ref	Comment/Dependencies
QA-F-040	Provide filters (tags, date, language), sort (relevance/recency), and pagination/infinite scroll	Functional	High	End Users, Admin	Filters persist in URL; empty state and no-results messaging. *Test:* TC-UI-01.	Requires front-end filter UI + API query parameters; ensure empty and no-results states are handled.
QA-F-041	Present canonical answer page with shareable URL, breadcrumb tags, and related questions.	Functional	Medium	End Users, Admin	OpenGraph meta present; related items computed by shared tags and co-clicks. *Test:* TC-UI-02.	Requires CMS-style front-end template + SEO meta; ensure related questions are computed correctly.

4.6 Audit and Compliance

Req Id	Requirements	Type	Priority	Source/Stakeholders	Acceptance/Test case ref	Comment/ Dependencies

QA-F-050	Record audit logs for auth, CRUD, publish, delete, and reindex actions.	Functional	High	Admin, Compliance Officers	Logs include who/when/what (old/new); immutable store. *Test:* TC-AUD-01.	Requires audit logging service + secure storage; ensure immutability and accessibility for audits.
QA-F-051	Export audit logs (CSV) for a date range and user.	Functional	Medium	Admin, Compliance Officers	Download produces complete, signed CSV (hash). *Test:* TC-AUD-02.	Requires export API + CSV generator; ensure only admins can access exports.

4.7 Analytics and Reporting

Req Id	Requirements	Type	Priority	Source/Stakeholders	Acceptance/Test case ref	Comment/Dependencies
QA-F-060	Admin dashboard metrics: searches/day, top queries, zero-result queries, CTR, helpful rate.	Functional	High	Admin, Analytics Team	Charts render; numbers match stored events. *Test:* TC-ANL-01.	Requires analytics DB + visualization charts; ensure data refresh in near real-time.

QA-F-061	Report builder: filter by tag/date; export CSV.	Functional	Medium	Admin, Analytics Team	Generated CSV matches on-screen aggregation. *Test:* TC-ANL-02.	Requires reporting engine + export module; ensure large reports are handled efficiently.

4.8 Reliability and Ops

Req Id	Requirements	Type	Priority	Source/Stakeholders	Acceptance/Test case ref	Comment/Dependencies
QA-F-070	Health checks: `/healthz` (liveness), `/readyz` (readiness: DB + index reachable).	Functional	High	DevOps, System Admin	Probes reflect dependency status; Kubernetes friendly. *Test:* TC-OPS-01.	Requires monitoring endpoints + integration with Kubernetes; ensure accurate dependency status.
QA-F-071	Scheduled backups of DB and index metadata; restore procedure documented and tested.	Functional	High	DevOps, System Admin	Restore into staging passes integrity checks. *Test:* TC-OPS-02.	Requires backup scheduler + secure storage; ensure documented restore steps tested regularly.

4.9 Security

Req Id	Requirements	Type	Priority	Source/Stakeholders	Acceptance/Test case ref	Comment/Dependencies
QA-F-080	Server-side validation and content sanitization (answers, tags).	Functional	High	Security Officer, Dev Team	XSS payloads neutralized; validation errors returned with fields. *Test:* TC-SEC-01.	Requires validation library + sanitizer; ensure strong test coverage for edge cases.
QA-F-081	Rate limit public APIs; lockout on brute-force login attempts.	Functional	High	Security Officer, Admin	Exceeding limits yields 429; repeated failed logins trigger temporary block. *Test:* TC-SEC-02.	Requires API gateway + rate-limiter; ensure configurable thresholds and monitoring.
QA-F-082	Permissions on export/import endpoints restricted to `admin`.	Functional	High	Admin, Security Officer	Non-admin receives 403; attempts logged. *Test:* TC-SEC-03.	Requires role-based access checks + audit logs; ensure attempts are logged for compliance.

5. Non-functional requirements (detailed)

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
FAQ-NF-001	The system shall return ranked search results within ≤ 2 seconds for 95% of queries under normal load.	Performance	High	Load test with 1000 queries shows 95% responses ≤ 2s.
FAQ-NF-002	The system shall ensure 99% uptime monthly, excluding scheduled maintenance	Reliability/Availability	High	Uptime monitoring confirms ≥ 99%.
FAQ-NF-003	The system shall use HTTPS with TLS 1.2+ for all network communications	Security/Compliance	High	Verified through SSL/TLS test tools
FAQ-NF-004	The system shall maintain logs of all admin CRUD operations with timestamps.	Auditability	Medium	Log review confirms all actions recorded.
FAQ-NF-005	The system shall provide average page load times under 3	Performance/Usability	Medium	Browser performance test confirms < 3s.

	seconds on standard 4G mobile networks.			

5.1. Security

5.1.1 Security Objectives

- Ensure that all admin and user communications are encrypted to protect data confidentiality.

- Prevent unauthorized access to CRUD operations by enforcing role-based authentication.

- Protect against malicious search inputs (e.g., injection attacks).

- Maintain audit logs for accountability and

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Measurement
FAQ-SR-001	TLS 1.2+ mandatory for all network communications.	Security	High	Verified with SSL/TLS scan tools
FAQ-SR-002	System shall require admin authentication before CRUD operations.	Security	High	Attempting CRUD without login must fail
FAQ-SR-003	System shall sanitize all user input in search to prevent injection attacks	Security	High	Penetration test shows no injection vulnerabilities
FAQ-SR-004	All admin CRUD operations shall be logged with timestamps.	Security	Medium	Logs reviewed to confirm completeness

FAQ-SR-005	Passwords shall be hashed using strong algorithms	Security	High	Security audit confirms no plaintext passwords

6. Quality attributes & Acceptance tests

Quality Attribute	Description	Acceptance Tests
Performance	The system shall return search results within ≤ 2 seconds for 95% of queries under normal load	Run 1000 search queries; 95% must return in ≤ 2s.
Reliability	The system shall provide 99% uptime monthly excluding scheduled maintenance.	Monitor uptime logs for 30 days; downtime < 1%.
Security	All communications encrypted with TLS 1.2+; CRUD restricted to authenticated admins.	Attempt unauthorized CRUD/search; system blocks & logs event.
Maintainability	The system shall use modular design (UI, API, indexing separated)	Code review verifies modularity; API endpoints documented.

7. System models and diagrams

7.1 UML Use-Case diagram

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Short	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
QA-F-001	Authenticate users & issue JWT tokens	4.1 Authentication & RBAC	Auth Module	TC-AUTH-01	N	Needs JWT library, secure password hashing, TLS.
QA-F-002	Enforce RBAC with roles (admin, editor, viewer)	4.1 Authentication & RBAC	Auth / RBAC	TC-AUTH-02	N	Requires role-permission mapping & authorization middleware.
QA-F-003	Support SSO via OAuth2/OIDC	4.1 Authentication & RBAC	Auth / SSO	TC-AUTH-03	N	Needs OAuth2/OIDC library + role mapping config.
QA-F-010	Create Q/A (question, answer, tags, status, owner, language, SEO fields)	4.2 Q/A Content Mgmt	Q/A CRUD API	TC-QA-01	N	Ensure mandatory fields & language support.
QA-F-011	Update Q/A with auto-versioning	4.2 Q/A Content Mgmt	Q/A CRUD API	TC-QA-02	N	Requires DB versioning & restore support.

QA-F-012	Soft delete/archive Q/A	4.2 Q/A Content Mgmt	Q/A CRUD API	TC-QA-03	N	Archived items excluded from search.
QA-F-013	Bulk import Q/A (CSV/JSON)	4.2 Q/A Content Mgmt	Q/A Import Module	TC-QA-04	N	Validate rows, error reporting.
QA-F-014	Tag/category management (create/merge/rename)	4.2 Q/A Content Mgmt	Tag Mgmt	TC-QA-05	N	Prevent orphan tags, reindex after merge.
QA-F-015	Publish workflow (draft → review → published)	4.2 Q/A Content Mgmt	Workflow Module	TC-QA-06	N	Clarify if review step mandatory.
QA-F-020	Index Q/A fields with boosts (Q > A > tags)	4.3 Search & Ranking	Indexing Engine	TC-SRCH-01	N	Requires boosted schema config.
QA-F-021	Keyword search ranked by relevance	4.3 Search & Ranking	Search API	TC-SRCH-02	N	Deterministic ranking validation.
QA-F-022	Support phrase queries, boolean ops, filters	4.3 Search & Ranking	Search API	TC-SRCH-03	N	Requires advanced query parser.
QA-F-023	Highlight matched terms in snippets	4.3 Search & Ranking	Search API / UI	TC-SRCH-04	N	Ensure safe HTML sanitization.

QA-F-024	Typo tolerance + stemming	4.3 Search & Ranking	Search API	TC-SRCH-05	N	Language-aware, configurable.
QA-F-025	Reindex on create/update/delete within 2s	4.3 Search & Ranking	Index Sync	TC-SRCH-06	N	Requires event-driven sync & manual fallback.
QA-F-030	Suggested answers (autocomplete)	4.4 Suggestions & Feedback	Search API / UI	TC-SUG-01	N	<150ms prefix queries.
QA-F-031	Capture feedback (helpful / not helpful)	4.4 Suggestions & Feedback	Feedback API / DB	TC-SUG-02	N	Requires feedback DB + analytics link.
QA-F-032	Adjust boosts from implicit feedback (learning-to-rank lite)	4.4 Suggestions & Feedback	Analytics + Ranking	TC-SUG-03	N	Requires CTR-based boosting.
QA-F-040	Filters, sorting, pagination/infinite scroll	4.5 End-User Search UI	Search UI	TC-UI-01	N	Handle empty/no-results states.
QA-F-041	Canonical answer page (URL, breadcrumbs, related Qs)	4.5 End-User Search UI	Search UI	TC-UI-02	N	Requires SEO metadata.
QA-F-050	Audit logs for CRUD, publish, delete, reindex	4.6 Audit & Compliance	Audit Service	TC-AUD-01	N	Logs must be immutable.

QA-F-051	Export audit logs (CSV)	4.6 Audit & Compliance	Audit Export Module	TC-AUD-02	N	Restricted to admins.
QA-F-060	Dashboard metrics (search/day, top queries, CTR, helpful rate)	4.7 Analytics & Reporting	Analytics Dashboard	TC-ANL-01	N	Charts must refresh in near real-time.
QA-F-061	Report builder with CSV export	4.7 Analytics & Reporting	Reporting Engine	TC-ANL-02	N	Handle large reports efficiently.
QA-F-070	Health checks (/healthz, /readyz)	4.8 Reliability & Ops	Monitoring Service	TC-OPS-01	N	Kubernetes-friendly.
QA-F-071	Scheduled backups & restore procedure	4.8 Reliability & Ops	Backup/Restore Service	TC-OPS-02	N	Must pass integrity tests.
QA-F-080	Server-side validation & sanitization	4.9 Security	Validation Service	TC-SEC-01	N	Prevent XSS & invalid input.
QA-F-081	API rate limiting & brute-force lockout	4.9 Security	API Gateway / Auth	TC-SEC-02	N	Configurable thresholds.
QA-F-082	Restrict export/import endpoints to admins	4.9 Security	API Gateway / RBAC	TC-SEC-03	N	Unauthorized attempts logged.
FAQ-NF-001	Return ranked results ≤ 2s (95% queries)	5. Non-functional	Search API	PerfTest-NF-01	N	Load test validation.

FAQ-NF-002	Ensure 99% uptime monthly	5. Non-functional	Infra / Ops	AvailTest-NF-01	N	Excludes scheduled downtime.
FAQ-NF-003	TLS 1.2+ mandatory	5. Non-functional / Sec.	Security Layer	SecTest-NF-01	N	Verified with SSL scan.
FAQ-NF-004	Maintain logs of all admin CRUD ops	5. Non-functional	Audit Service	AuditTest-NF-01	N	Confirm log completeness.
FAQ-NF-005	Avg page load ≤ 3s on 4G	5. Non-functional	Frontend UI	UXTest-NF-01	N	Confirm via browser perf test.
FAQ-SR-001	TLS 1.2+ enforced	5.1 Security Requirements	Security Layer	SecTest-SR-01	N	Verified with SSL/TLS scans.
FAQ-SR-002	Require admin auth before CRUD	5.1 Security Requirements	Auth Module	SecTest-SR-02	N	Unauthorized CRUD blocked.
FAQ-SR-003	Sanitize user input to prevent injection	5.1 Security Requirements	Validation Layer	SecTest-SR-03	N	Pen-test validation.
FAQ-SR-004	Log all admin CRUD ops with timestamps	5.1 Security Requirements	Audit Service	SecTest-SR-04	N	Log review confirms completeness.
FAQ-SR-005	Hash passwords using strong algorithm	5.1 Security Requirements	Auth Module	SecTest-SR-05	N	Verified during security audit.